



REACH CAD

Using the Block Method for Pattern Drafting in REACH CAD

Learning Guide

Using the Block Method for Pattern Drafting in REACH CAD

The block method in REACH CAD is a systematic approach to pattern drafting, utilizing foundational blocks as templates to create consistent and accurate garment patterns. These blocks serve as reusable base patterns, streamlining design, ensuring uniformity, and facilitating efficient modifications. This guide details the workflow for using the block method in the Pattern Designing and Grading System (PDS) module, based on the official REACH CAD user manual and industry best practices.



01

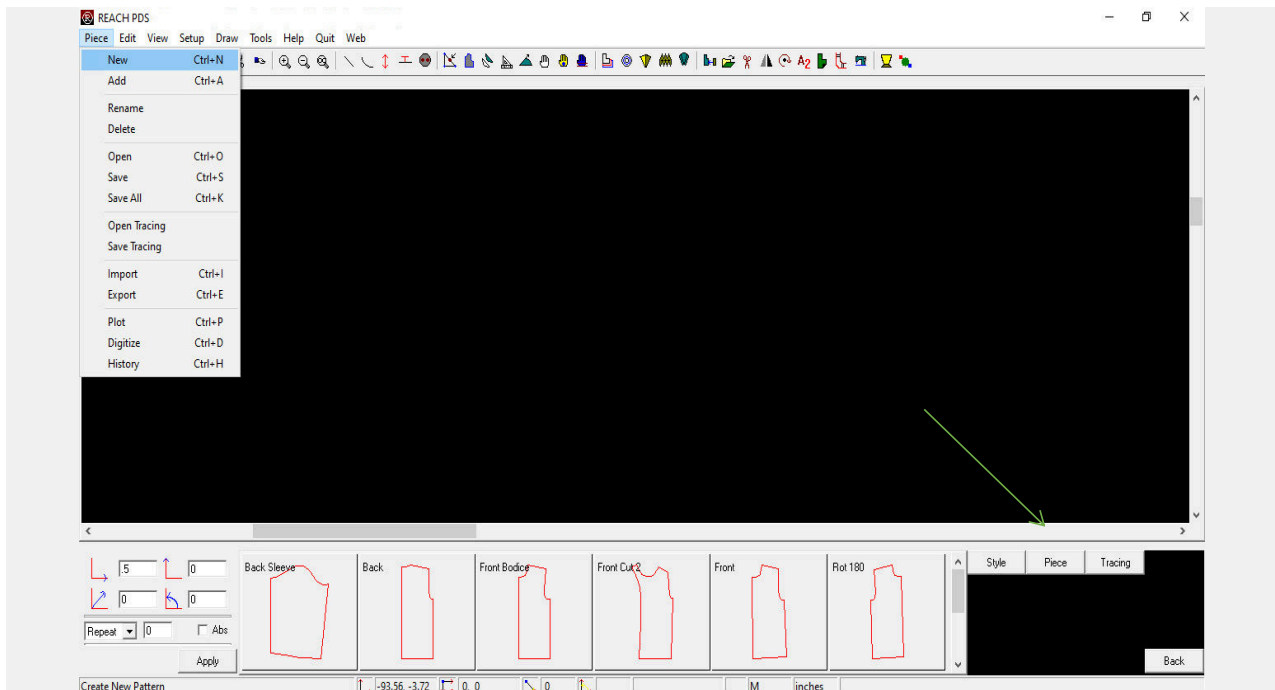
Creating a New Block

1.Creating a New Block

A. **Purpose:** Establish a foundational template (block) for a specific garment component (e.g., bodice, sleeve, skirt) to serve as the starting point for pattern development.

B. How to Create a Block:

- Navigate to PIECE > New > Piece (shortcut: Ctrl + N) in the REACH PDS module.
- In the dialog box, select Piece to initiate block creation.



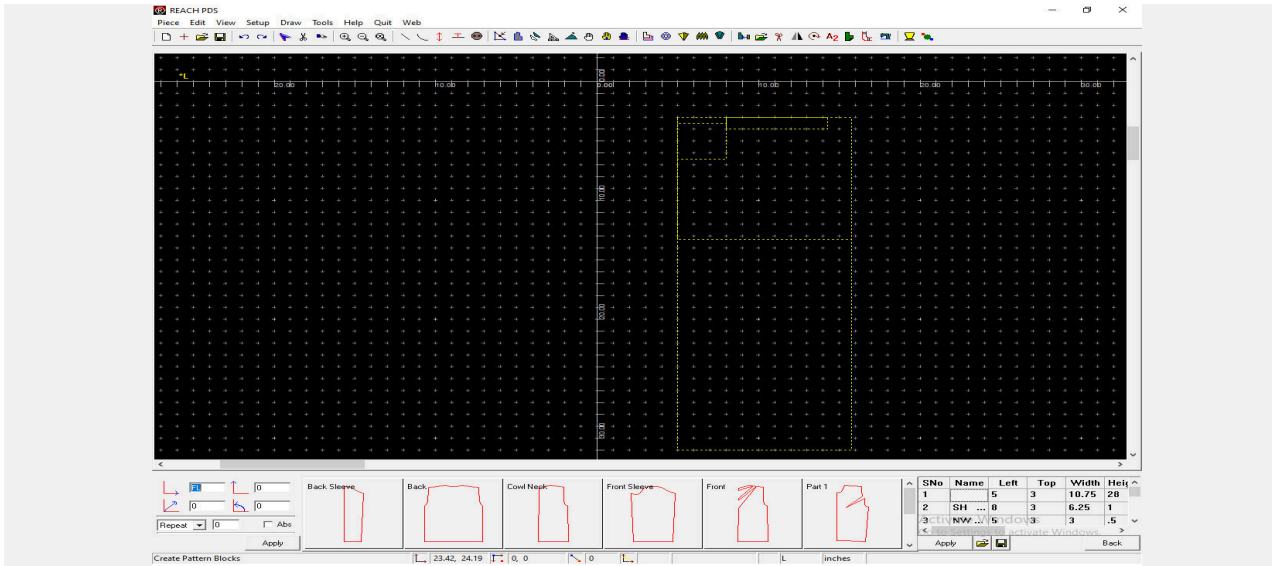
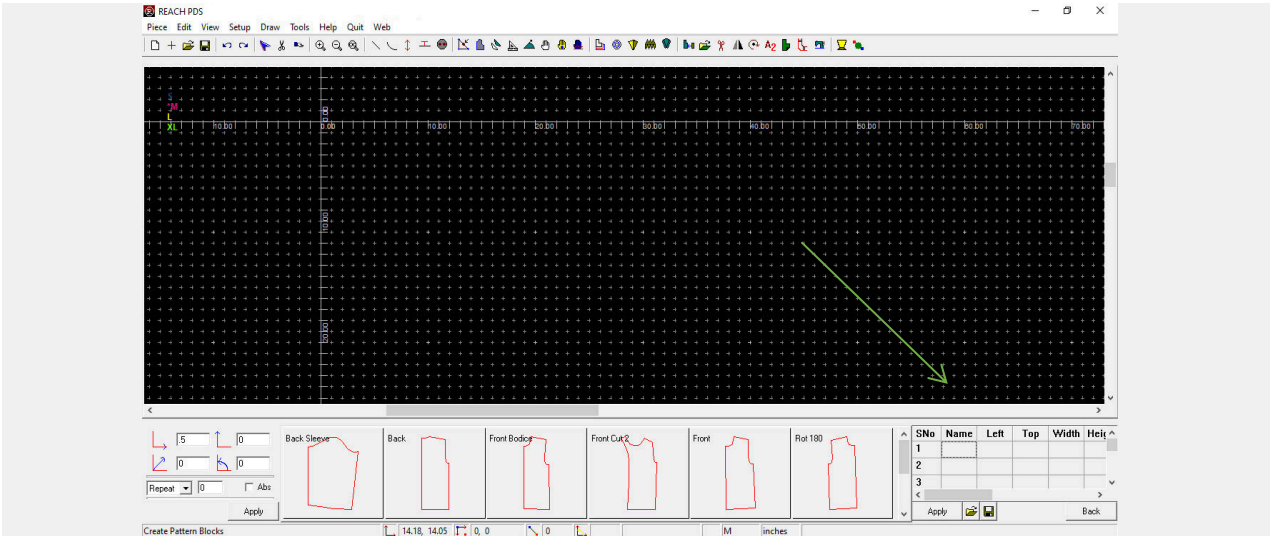
C. Define block parameters:

- Name: Assign a clear, descriptive name (e.g., "Bodice_Block_Front").
- Dimensions: Specify Left, Top, Width, and Height to set the block's size and position in the workspace.

D. Click Apply to create the block and display it in the working area.



Learning Guide



REACH CAD



02

Drafting the Base Pattern

2. Drafting the Base Pattern

A. **Purpose:** Develop the detailed pattern outline within the block's boundaries, forming the core shape of the garment component.

B. How to Draft:

a. Use drawing tools in the Draw menu:

i. Line/Curve: Draft straight lines for seams (e.g., side seams) and curves for contoured edges (e.g., armholes, necklines).

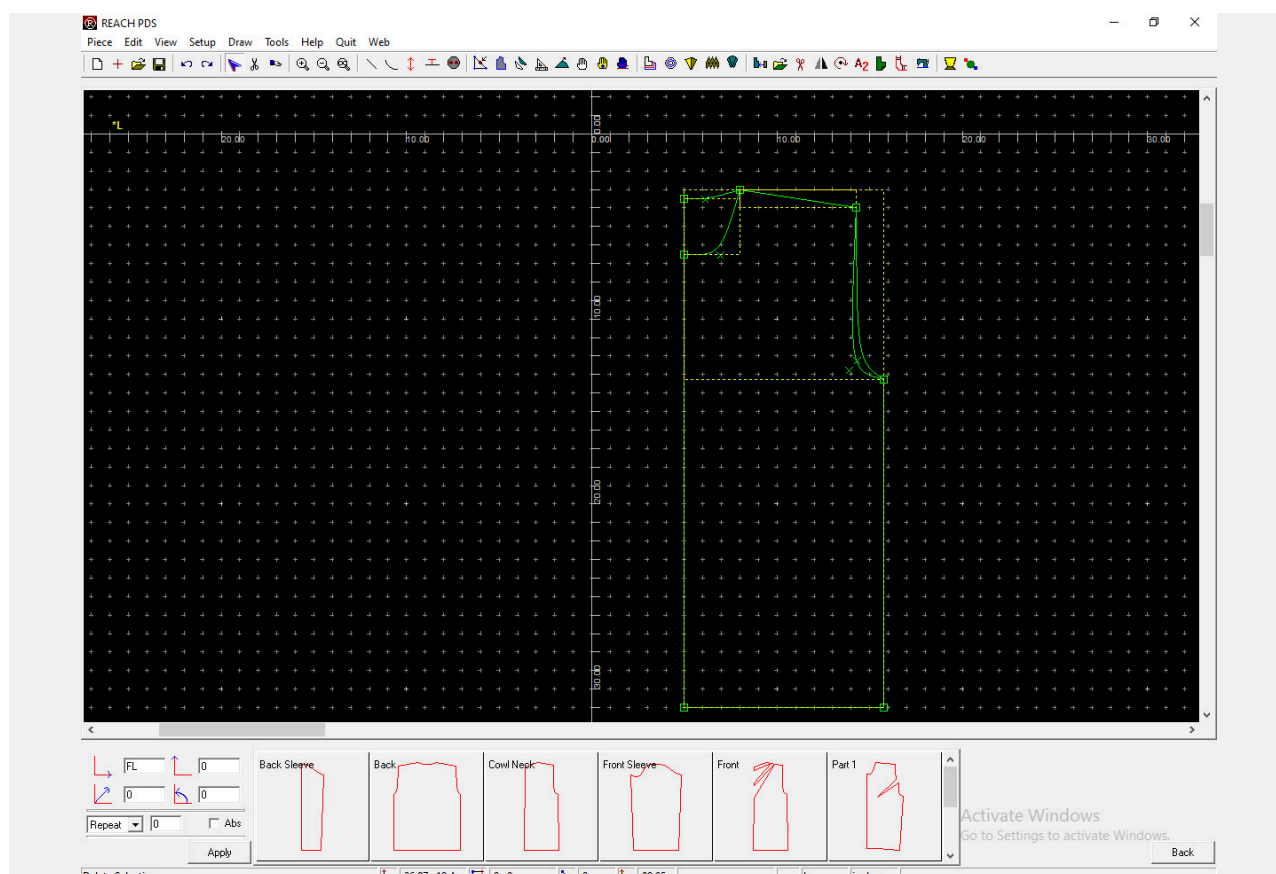
b. Refine the pattern using editing tools:

i. Move Pin (Ctrl + F9): Adjust individual points for precise shaping.

ii. Trim (Ctrl + T): Remove overlapping or unnecessary lines.

iii. Duplicate (Shift + F8): Create symmetrical elements (e.g., left/right fronts).

c. Ensure the pattern adheres to the block's dimensions for consistency.



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03

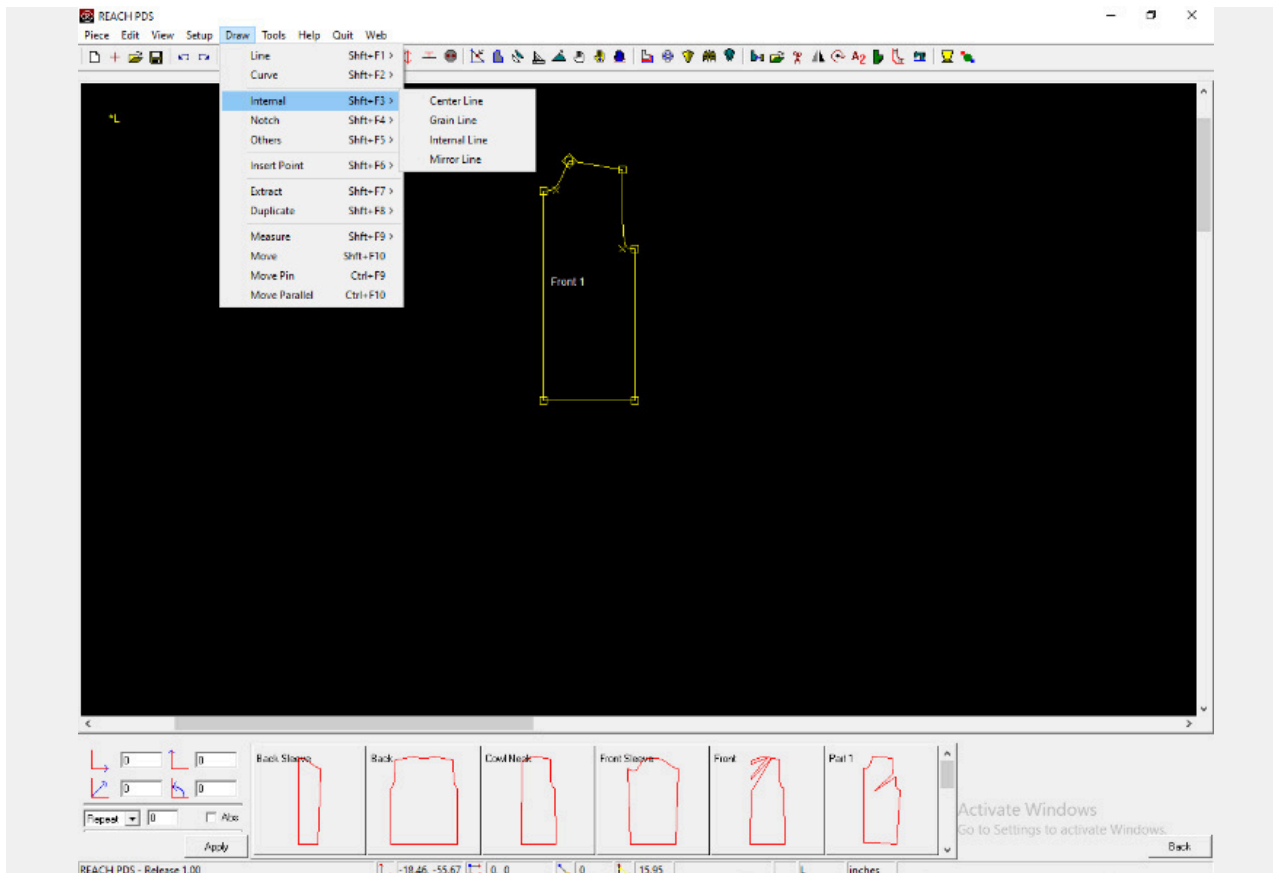
Adding Internals and Notches

3. Adding Internals and Notches

A. **Purpose:** Incorporate critical construction marks to guide fabric alignment and assembly.

B. How to Add:

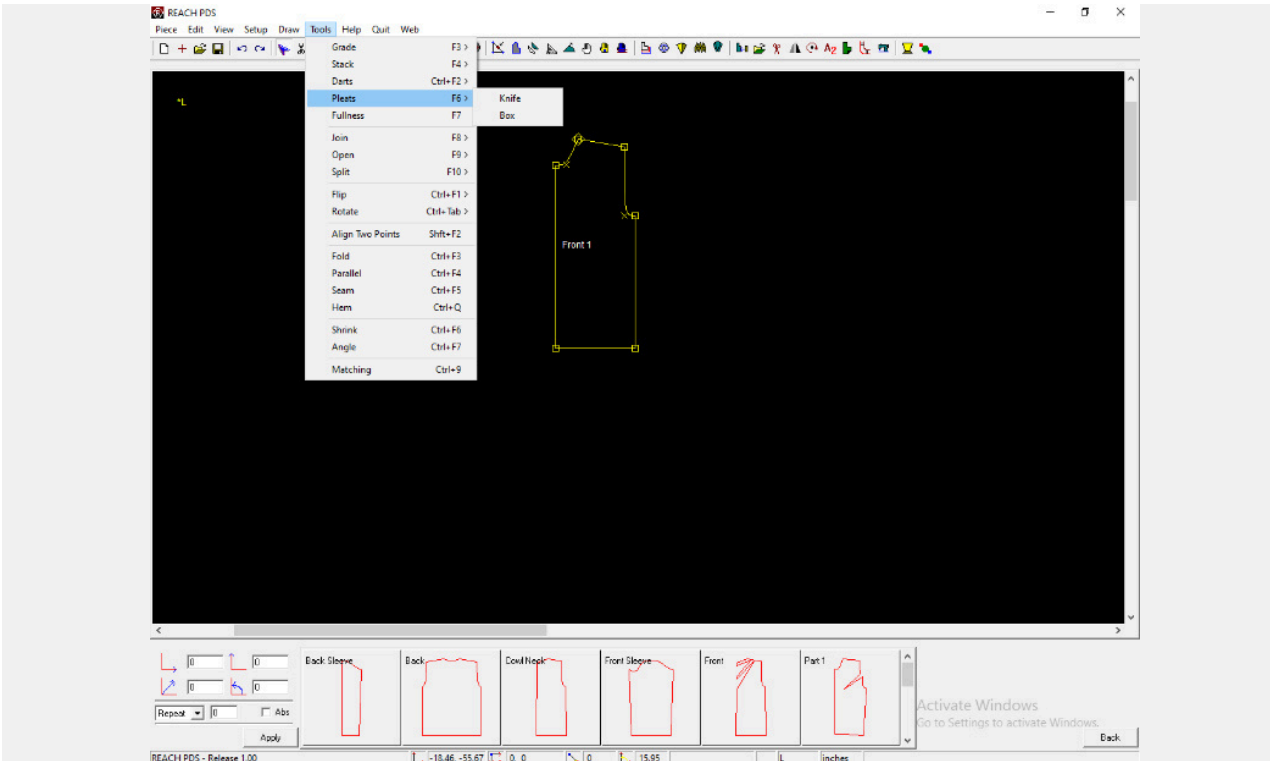
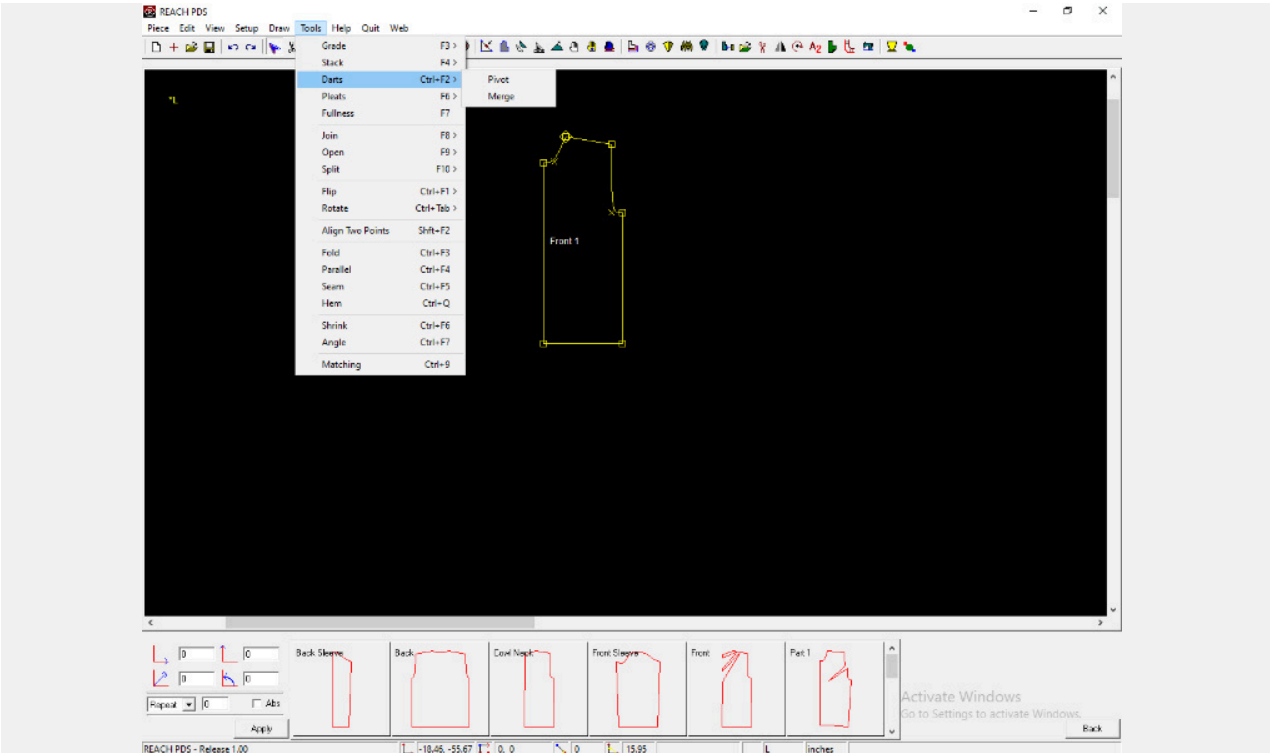
a. Grain Lines: Use Draw > Internal > Grain Line to indicate fabric direction.



b. Darts/Pleats: Draft using Draw > Internal > Dart Line or Internal Line for precise placement.



Learning Guide

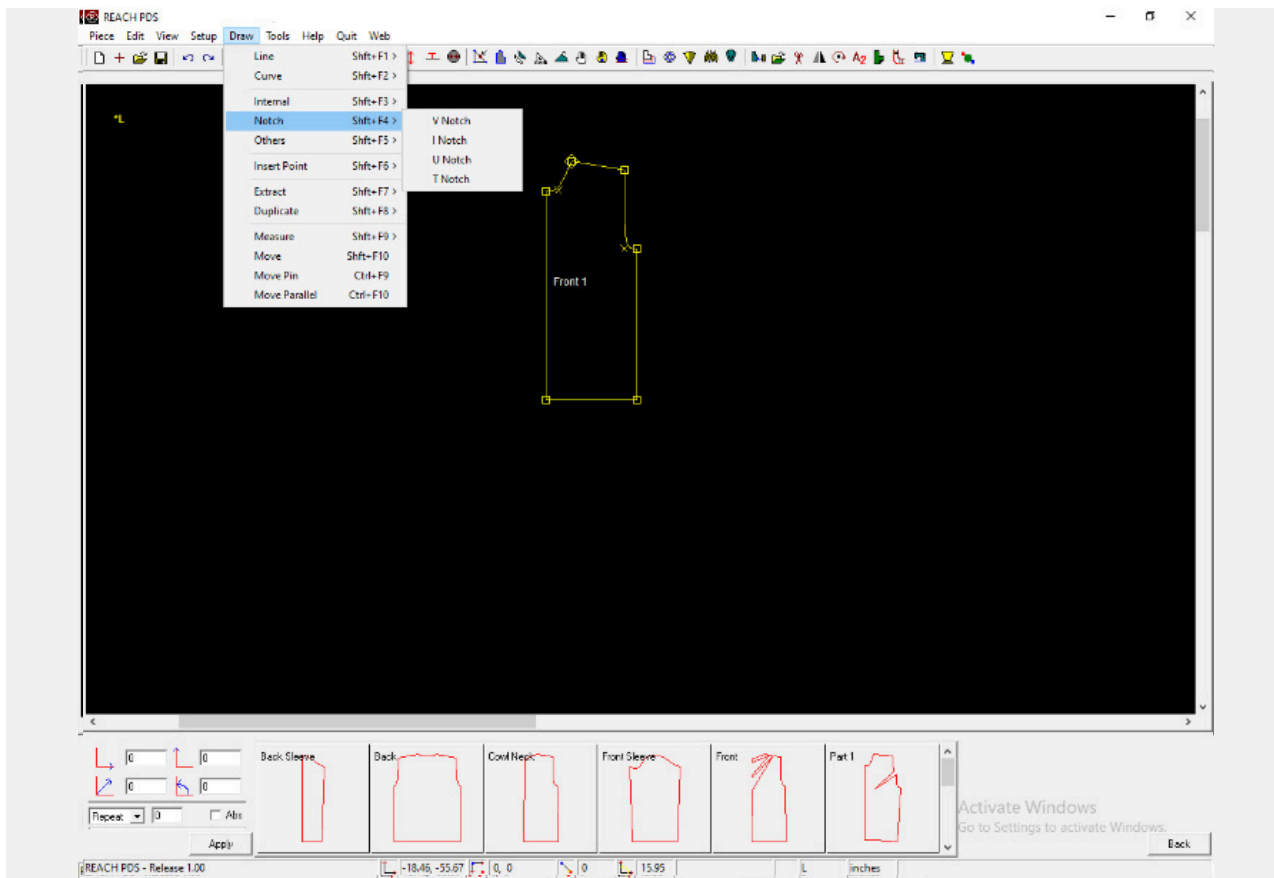


REACH CAD



Learning Guide

c. Notches: Add V, I, U, or T notches via Draw > Notch to mark alignment points for seams or folds.



d. Verify alignment of internals with the pattern outline to ensure production accuracy.

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04

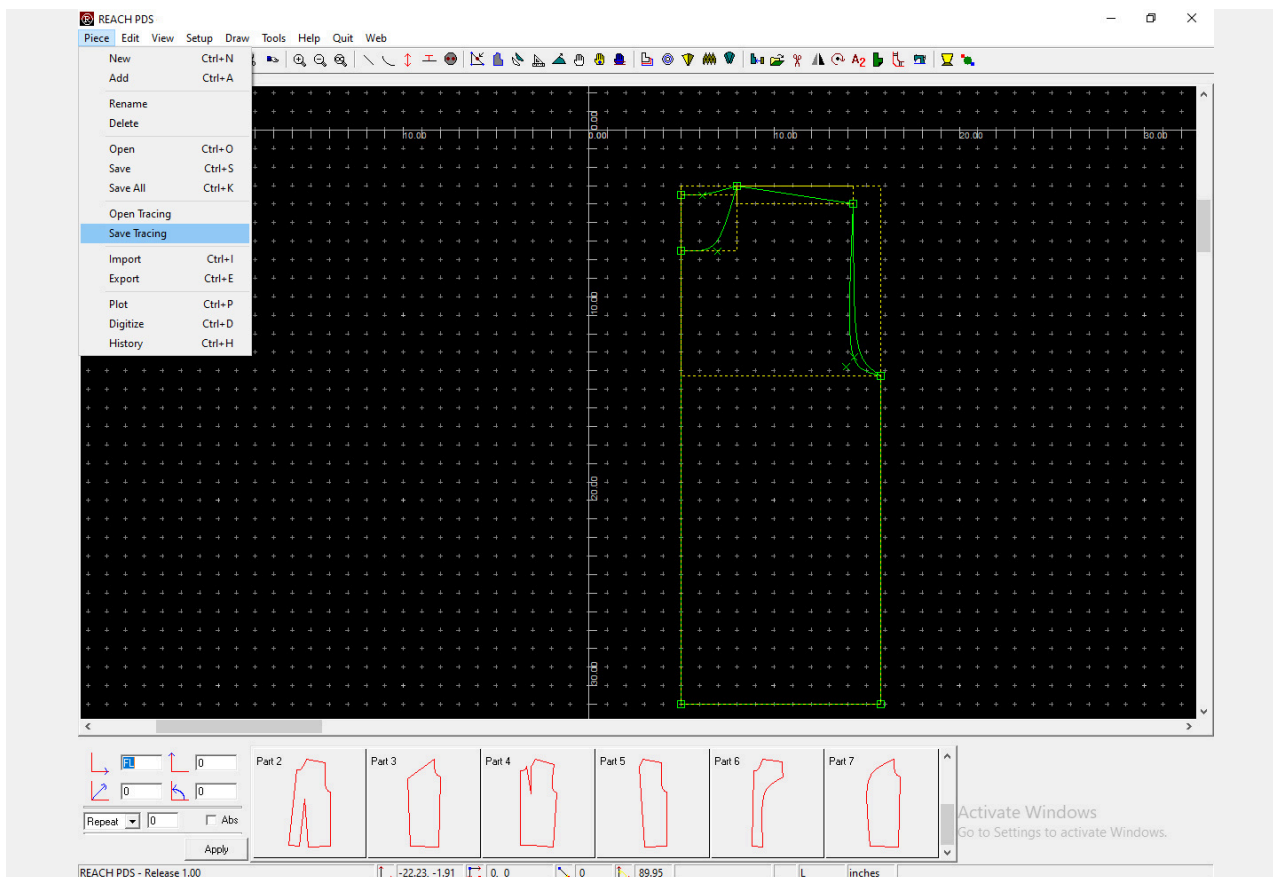
Saving and Reusing Blocks

4. Saving and Reusing Blocks

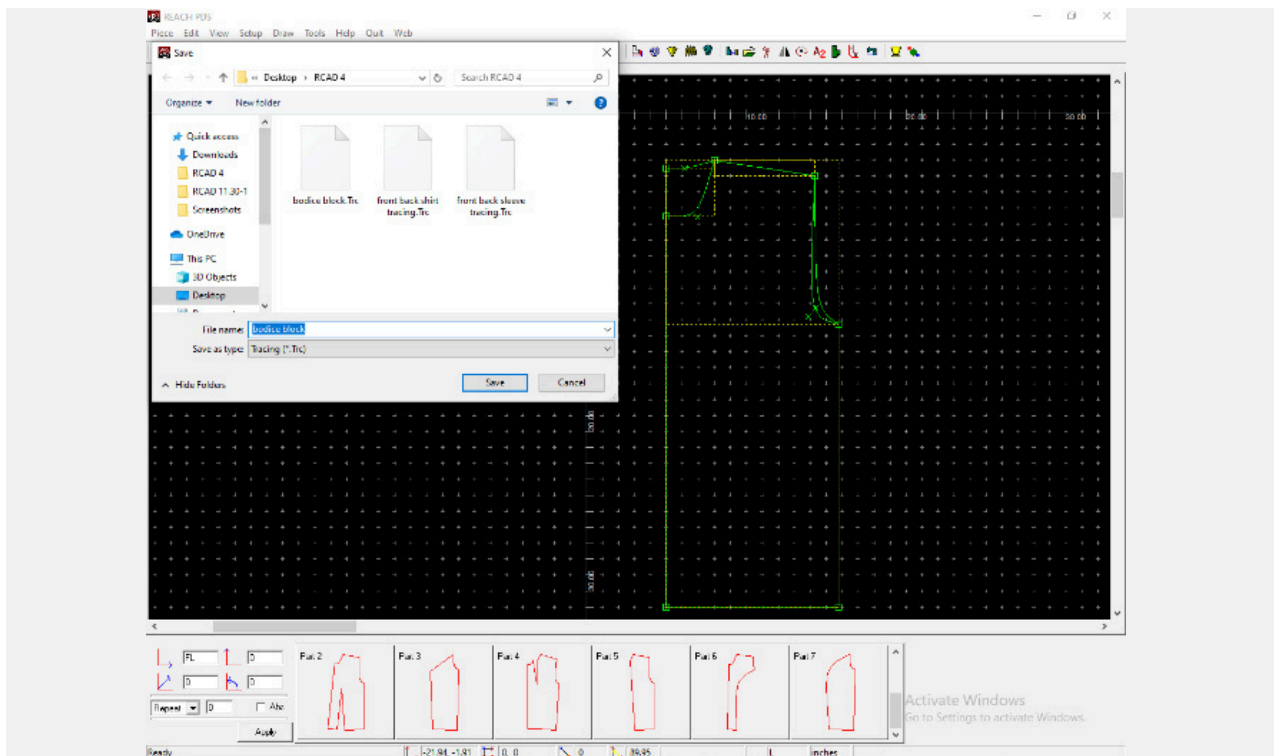
A. **Purpose:** Save blocks as reusable templates to streamline future pattern development for similar styles.

B. How to Save and Reuse:

- Save the block via **PIECE > Save** (Ctrl + S) with a unique name for easy retrieval.
- Access saved blocks through **SETUP > Blocks** (Ctrl + 7) to load them into new projects.
- Reuse blocks for variations (e.g., modify a "Basic Sleeve" block for a puff sleeve design).



Learning Guide



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05

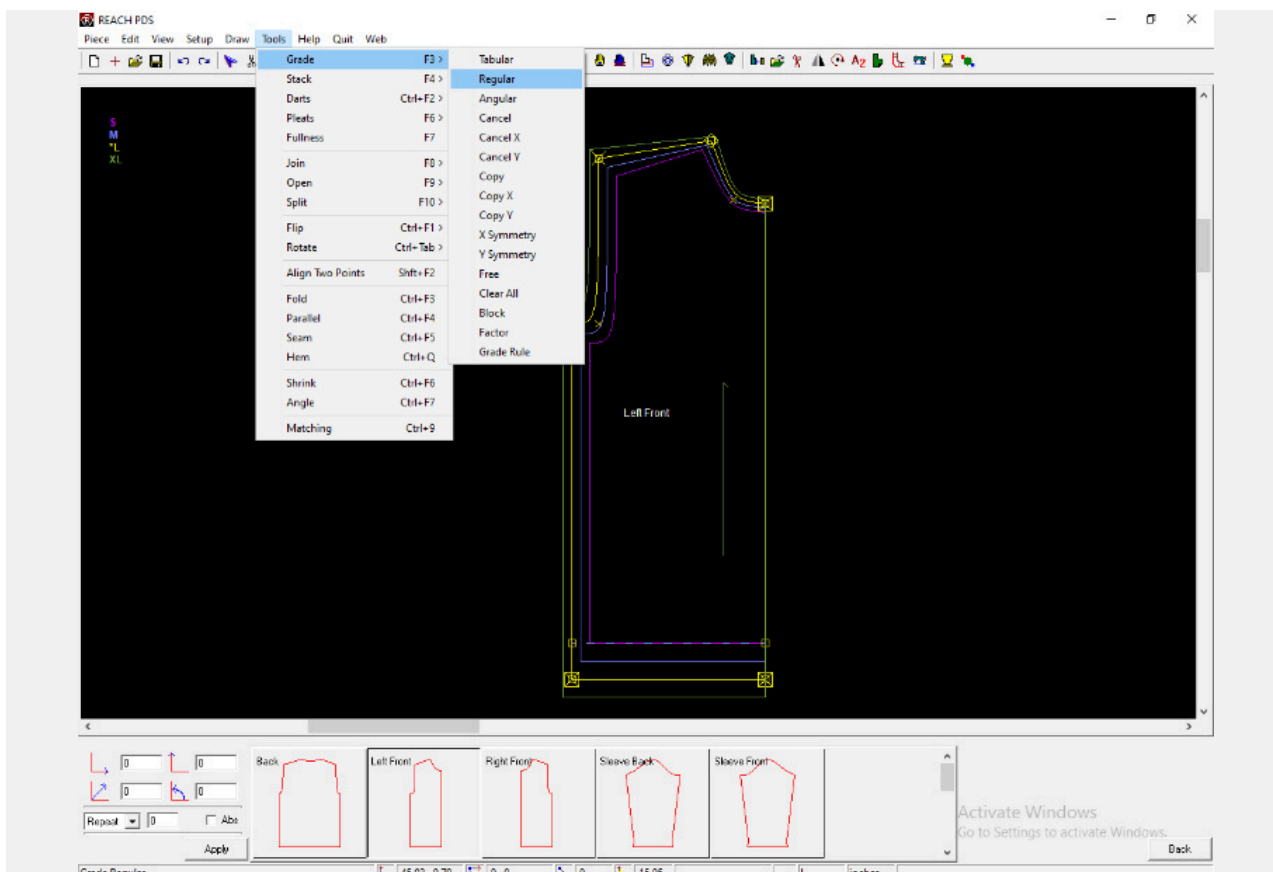
Grading from Blocks

5. Grading from Blocks

A. **Purpose:** Scale the block-based pattern across a range of sizes while maintaining its core structure.

B. How to Grade:

- a. Define the size table in SETUP > Size (Ctrl + 1), listing sizes (e.g., S, M, L, XL).
- b. Apply grading rules via Tools > Grade > Tabular or Regular
 - i. Enter X/Y increments for each size relative to the base size (e.g., +2 cm chest width per size)
- c. Use View > Stack/Nest to verify graded patterns for smoothness and alignment.
- d. Save graded blocks for use in marker planning.



Workflow Summary

Step	Action	Purpose
Create Block	PIECE > New > Piece (Ctrl + N); set name, dimensions	Establish template for pattern drafting
Draft Base Pattern	Use Draw tools (Line/Curve); refine with Move Pin, Trim	Develop accurate pattern outline
Add Markings	Insert grain lines, darts, notches via Draw > Internal/Notch	Ensure proper fabric alignment and assembly
Save/Reuse	PIECE > Save; access via SETUP > Blocks (Ctrl + 7)	Enable reuse for similar styles
Grade	SETUP > Size; Tools > Grade; apply increments	Scale block for multiple sizes

Best Practices

- A. **Standardize Blocks:** Create a library of standard blocks (e.g., bodice, sleeve, skirt) for consistent pattern development across styles.
- B. **Layered Workflow:** Draft the outer contour first, then add internals (darts, notches) to maintain clarity.
- C. **Validate Grading:** Use View > Stack/Nest to check graded patterns for accuracy and alignment.
- D. **Organize Block Library:** Use descriptive names and organize blocks by garment type or season in SETUP > Blocks.
- E. **Save Frequently:** Save blocks and patterns (Ctrl + S) to prevent data loss during drafting



In summary:

The block method in REACH CAD provides a structured, efficient approach to pattern drafting by leveraging reusable templates. By creating, drafting, and grading blocks and incorporating internals and notches, users can ensure consistent, accurate patterns that streamline design iterations and production workflows.



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